

Phase 5: Proof Trees

Example 1 : Simple Implication

$$\frac{\frac{\frac{\lceil p \wedge q \rceil^{[1]}}{p \wedge q} [\wedge\text{-elim-2}]}{q} [\Rightarrow\text{-intro}^{[1]}]}{p \wedge q \Rightarrow q}$$

Example 2 : With Sibling Premises

$$\frac{\frac{\frac{\lceil p \wedge (p \Rightarrow q) \rceil^{[1]}}{p} [\wedge\text{-elim-1}]}{\frac{q}{p \wedge q} [\wedge\text{-intro}]} [\Rightarrow\text{-elim}]}{p \wedge (p \Rightarrow q) \Rightarrow (p \wedge q)} [\Rightarrow\text{-intro}^{[1]}]$$

Example 3 : Distribution with Cases

$$\frac{\frac{\frac{\frac{\neg}{p} [\text{from above}]}{q} [\text{from case}]}{p \wedge q} [\wedge\text{-intro}]}{(p \wedge q) \vee (p \wedge r)} [\vee\text{-intro-1}]}{\frac{\frac{\frac{\neg}{p} [\text{from above}]}{r} [\text{from case}]}{p \wedge r} [\wedge\text{-intro}]} [\vee\text{-intro-2}]}{(p \wedge q) \vee (p \wedge r)} [\vee\text{-elim}]}{\frac{(p \wedge q) \vee (p \wedge r)}{p \wedge (q \vee r) \Rightarrow (p \wedge q) \vee (p \wedge r)} [\Rightarrow\text{-intro}^{[1]}]}$$

Example 4 : Modus Tollens

$$\frac{\frac{\frac{\lceil (p \Rightarrow q) \wedge \neg q \rceil^{[1]}}{p \Rightarrow q} [\wedge\text{-elim-1}]}{\frac{\lceil (p \Rightarrow q) \wedge \neg q \rceil^{[1]}}{\neg q} [\wedge\text{-elim-2}]} [\Rightarrow\text{-elim}]}{\frac{\frac{\lceil p \rceil^{[2]}}{q} [\Rightarrow\text{-elim}]}{\frac{q}{false} [\text{contradiction}]} [\neg\text{negation-intro}^{[2]}]} [\Rightarrow\text{-intro}^{[1]}]}{(p \Rightarrow q) \wedge \neg q \Rightarrow \neg p}$$

Example 5 : Solution 18 Implication to Disjunction

$$\frac{\frac{\frac{\lceil p \Rightarrow q \rceil^{[1]}}{p \vee \neg p} [\text{excluded middle}]}{\frac{\lceil p \rceil^{[2]}}{q} [\Rightarrow\text{-elim}]} [\vee\text{-intro-2}]}{\frac{\frac{\lceil \neg p \rceil^{[2]}}{\neg p \vee q} [\vee\text{-intro-1}]}{\neg p \vee q} [\vee\text{-elim}^{[2]}]} [\Rightarrow\text{-intro}^{[1]}]}{(p \Rightarrow q) \Rightarrow (\neg p \vee q)}$$